5 Practical Generative AI Use Cases for Public Sector

Concept Note — Assignment #6: Introduction to Generative AI Applications

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Introduction to Generative AI

Generative AI refers to a category of artificial intelligence systems that can create new content, such as audio, video, text, or other data, rather than only analyzing or classifying existing data. Generative AI models learn the structure and patterns of large amounts of data and use that understanding to produce new, human-like output. The most common type of generative AI is the LLM (Large Language Model), such as GPT, Claude, and Gemini. These models are trained on vast amounts of data to understand patterns, styles, and facts, so when a prompt is given, an LLM generates a relevant response by predicting the most likely sequence of words.

In the public sector specifically, generative AI has practical potential in areas such as citizen communication, report summarization, meeting documentation, and policy Q&A — tasks that are repetitive, time-consuming, and dependent on accurate language and formal tone. This concept note presents five practical Generative AI use cases relevant to a public-sector office such as PITB. Each use case is explained through its problem statement, target user, input, AI process, expected output, limitations, and possible improvements, followed by the actual prompt used and the real, tested output received from an LLM.

Responsible AI and Limitations

Across all five use cases, a few responsible-AI principles apply consistently. First, AI-generated content (emails, summaries, minutes, answers, or ideas) should always be reviewed by a human before being sent, published, or acted upon, especially where it affects a citizen's request or an official record. Second, systems that process citizen data (such as the Email Assistant or Document Q&A Assistant) must handle personal information responsibly and avoid storing or exposing sensitive details unnecessarily. Third, users should be made aware when they are interacting with AI-generated content rather than a human response. Finally, every use case in this note carries a general limitation of generative AI: the model can only work with the information it is given, cannot verify facts outside that input, and may occasionally produce plausible-sounding but incorrect output (hallucination) if the prompt does not explicitly guard against it.

Underlying AI Techniques Used

While each use case is framed around text generation, several also rely on other core Generative AI techniques. The Email Assistant and Document Q&A Assistant depend on **information extraction** — pulling specific facts (a fact from a citizen's email, or an answer from a policy document) out of unstructured text. The Document Q&A Assistant also relies on a form of **classification**, since the model must judge whether the requested information is present in the document at all before answering. The Report Summarizer and Meeting Notes Generator are primarily **summarization** tasks, condensing a

longer input into a shorter, structured one without losing key meaning.

Note: Each use case below includes an Evaluation Note explaining the prompt design (clarity, role, constraints, and format) and the real Actual Tested Output received when the prompt was run in an LLM.

Use Case 1: AI Email Assistant

Problem Statement: Government office staff spend a lot of time writing the same type of email repeatedly, such as replying to citizens' complaints or requests for documents. This takes time and slows down work.

Target User: Administrative staff across various government departments.

Input: The email received from a citizen, along with the key information that should be included in the reply, such as processing timelines or confirmation that a request has been received.

AI Process: The AI reads the citizen's email and the necessary information, then writes a professional email in the proper format.

Expected Output: A ready-to-send professional email with correct tone and no missing facts.

Limitations: The AI may not know current policies unless they are provided, and it may respond too generally to legal or official matters, so a person must review it before sending.

Possible Improvement: Connect the AI to the department's latest policy documents so it can always provide correct, up-to-date information.

Evaluation Note: This prompt succeeds because it defines a clear **role** (Ministry of Education), a specific **scenario** (scholarship delay), explicit **constraints** (under 150 words), and a defined **format** (greeting, body, closing). Comparing actual to expected: the real output matched the word limit, tone, and format exactly, confirming the prompt's constraints were both necessary and sufficient.

Tested Prompt:

“Write a formal reply email from the Ministry of Education to a student who is asking why their scholarship has not yet been granted, even though they applied a few months ago. Explain that the process takes time because the submitted documents must first be verified, and then the Ministry needs to receive confirmation from the student's university before the scholarship can be approved. Apologize for the delay, reassure the student that their application is being processed, and ask them to wait patiently. Keep the email under 150 words. Use a professional tone with a proper greeting, closing, and official email format.”

Actual Tested Output (real LLM response):

Dear Student,

Thank you for contacting the Ministry of Education regarding your scholarship application.

We understand your concern about the delay and sincerely apologize for the inconvenience. Please note that the scholarship approval process requires careful verification of all submitted documents. In addition, the Ministry must receive official confirmation from your university before the application can be finalized and approved.

We would like to assure you that your application is currently being processed, and every effort is being made to complete the review as soon as possible.

We kindly ask for your patience and understanding while this process is completed. If any additional information is required, we will contact you promptly.

Yours sincerely,
Scholarship Support Team
Ministry of Education

Use Case 2: AI Report Summarizer

Problem Statement: Government departments receive long technical or statistical reports but rarely have time to read them in full before meetings, risking uninformed decisions.

Target User: Senior officials and department heads in public-sector offices.

Input: A long report, such as a quarterly budget report.

AI Process: The AI reads the report and produces an executive summary with proper headings, sections, and page numbers.

Expected Output: A one-page summary with proper headings, sections, and page numbers.

Limitations: The AI may not understand charts and tables and may misinterpret them if formatting is lost in plain text; formatting can affect accuracy.

Possible Improvement: If the AI is uncertain about any number, text, or table, it should explicitly flag this instead of guessing.

Evaluation Note: The prompt's **constraints** (400-word limit, page numbers for every statistic) and **format** instruction (headings per department, ending conclusion) produced a well-organized, department-by-department summary as expected. The real output correctly cited page numbers for every statistic, showing the constraint was followed precisely rather than approximated.

Tested Prompt:

“Summarize this quarterly budget report. Organize the summary with headings for each department. Include the page number for every key statistic. Keep the summary under 400 words and end it with a short conclusion.”

Actual Tested Output (real LLM response):

Executive Overview
The Department of Education used 87% of its PKR 450,000,000 quarterly budget, spending PKR 391,500,000 and leaving PKR 58,500,000 unspent (Page 1). Overall spending was strongest in school operations, while infrastructure projects were delayed due to contractor procurement, and teacher training exceeded budget because of an unplanned digital-literacy initiative (Page 1).

School Operations — Allocated: PKR 180,000,000; Spent: PKR 168,300,000; Utilization: 93.5% (Page 2). Remained close to budget, reflecting stable spending on salaries, utilities, and school supplies.

Teacher Training & Development — Allocated: PKR 70,000,000; Spent: PKR 76,200,000; Utilization: 108.9% (Page 2). Spending exceeded allocation by PKR 6,200,000 due to a digital-literacy program reaching 1,200 teachers (Pages 2–3).

Infrastructure & Maintenance — Allocated: PKR 140,000,000; Spent: PKR 98,500,000; Utilization: 70.4% (Page 2). Largest variance, underspending by PKR 41,500,000 due to delayed contractor agreements (Pages 2 & 4).

Curriculum & Examinations — Allocated: PKR 60,000,000; Spent: PKR 48,500,000; Utilization: 80.8% (Page 2). Lower textbook printing costs led to moderate underspending.

Conclusion: The department demonstrated sound financial management with 87% budget utilization (Page 1). School Operations remained on target, Teacher Training exceeded expectations due to an approved initiative, while Infrastructure & Maintenance experienced significant delays that should be addressed before Q3 to improve budget execution.

Use Case 3: AI Meeting Notes Generator

Problem Statement: Public meetings, whether held in person or online (such as internal briefings), often go undocumented, making it hard to track decisions and follow-ups later.

Target User: Secretaries and office administrators.

Input: Input can be rough meeting notes, a written transcript, or a transcript generated from a recorded online meeting.

AI Process: The AI organizes the meeting transcript into structured minutes, covering topics discussed and decisions made. If a recording is provided, the audio is first converted to text using speech-to-text transcription, and the resulting transcript is then organized by the LLM into structured minutes.

Expected Output: A formatted minutes document with clear sections and a list of action items.

Limitations: If speaker names are not mentioned in the transcript, the AI cannot verify who said what. It may also misinterpret jokes or informal conversation, and speech-to-text transcription may introduce errors caused by background noise.

Possible Improvement: Use meeting notes or transcripts with clearly labeled speaker names. The AI could also be connected to online meeting platforms such as Zoom so it can automatically generate transcripts.

Evaluation Note: The **format** instructions (title, date, attendees, topics, decisions, action items with owner/deadline) map directly onto real minutes structure, and the **constraint** "do not add information not in the transcript" reduces hallucination risk. The real output not only matched this structure but also added a per-attendee summary section, showing the model correctly interpreted the "summary of important points by each attendee" instruction from the prompt.

Tested Prompt:

“Read this meeting transcript and convert it into formal meeting minutes. Include the meeting title (if available), date (if mentioned), attendees, topics discussed, key decisions, action items with the responsible person and deadline (if mentioned), and a summary of the important points discussed by each attendee. Use professional English, organize the information under clear headings, and present the important points in bullet form. Do not add any information that is not mentioned in the transcript.”

Actual Tested Output (real LLM response):

Meeting Title: Internal Briefing – Education Department (Zonal Office)
Date: July 20, 2026 **Time:** 11:00 AM – 11:45 AM
Attendees: M. Rizwan – Zonal Director; S. Fatima – Admin Officer; A. Noman – Infrastructure Coordinator; H. Bilal – Finance Officer

Topics Discussed: School renovation project updates; safety concerns at Government Girls School, Sector 9; teacher training budget overspend; quarterly report documentation; vacant teaching positions and recruitment planning; next meeting schedule.

Key Decisions: Delayed renovation projects to be finalized as soon as possible; boundary wall issue added to the priority list; teacher training overspend to be documented in the quarterly report; recruitment planning to be a priority next month; team to reconvene same time next week.

Action Items:
• Follow up with contractor and finalize renovation agreement — A. Noman — By Friday
• Include boundary wall issue in this week's site visit — A. Noman — This week's site visit
• Prepare note on teacher training overspend for quarterly report — H. Bilal — By Wednesday
• Draft recruitment timeline and share with Zonal Director — S. Fatima — End of next week

Summary by Attendee included individual bullet-point summaries for each of the four attendees, covering their specific contributions (e.g., Noman's report on contractor delays, Fatima's report on vacant posts, Bilal's report on the budget overspend).

Use Case 4: AI Document Q&A; Assistant

Problem Statement: Government staff and citizens often need quick answers from lengthy policy documents, but searching manually is slow and error-prone.

Target User: Citizens and government staff.

Input: A policy document and a specific question.

AI Process: The AI searches the document and answers the question using only the information available within it.

Expected Output: A direct, accurate answer with a reference to where in the document it was found.

Limitations: It may guess an answer if it is not present in the document, and very large documents may be difficult to process fully.

Possible Improvement: The AI should clearly state that a piece of information is not available in the document instead of guessing.

Evaluation Note: Restricting the AI to answer only from the provided document is the key **constraint** preventing hallucination, and the **format** requirement (bullet points) made the recommendations easy to scan. The real output correctly extracted the exact utilization figure and all three recommendations directly from the source document with no invented content.

Tested Prompt:

“Read this attached document and answer this question: What was the total utilization rate this quarter, and what are the department's recommendations? List the recommendations in bullet points.”

Actual Tested Output (real LLM response):

Total Utilization Rate: 87% of the allocated quarterly budget was utilized.

Department Recommendations:
• Expedite the contractor procurement process for pending renovation projects.
• Begin recruitment planning for vacant teaching positions ahead of the next academic year.
• Introduce a contingency allocation in future quarterly budgets to support unplanned but high-priority training initiatives without impacting other budget areas.

Use Case 5: AI Idea Generator for Public Services

Problem Statement: Government offices often need new ideas to improve public services but may not have a proper brainstorming process.

Target User: Policy planners and innovation teams in government departments.

Input: A description of the problem that needs to be resolved, such as long queues at a passport office.

AI Process: The AI generates multiple improvement ideas and explains how practical each one is.

Expected Output: A list of 5-7 useful ideas with simple notes about cost and implementation.

Limitations: Ideas may be too generic. The AI does not know the actual budget, so it might suggest ideas that don't fit financially, or ideas that have already been tried.

Possible Improvement: Feed the AI historical data on past initiatives so it avoids repeating solutions that already failed.

Evaluation Note: The prompt's **constraint** (5 ideas, one-line explanation each, cost/difficulty tag) forced concise, comparable output. The real output returned exactly 5 ideas formatted as a clean table with cost and implementation difficulty for each, closely matching the expected output with no extra filler content.

Tested Prompt:

“Suggest 5 simple and practical ideas to improve waste collection and cleanliness in a city. For each idea, write one short explanation and mention whether it is low, medium, or high cost and whether it is easy or difficult to implement.”

Actual Tested Output (real LLM response):

1. **Install more public trash bins** — Place additional bins in busy streets, parks, and markets to reduce littering. Cost: Low. Implementation: Easy.
2. **Increase waste collection frequency** — Collect garbage more often in high-population areas to prevent buildup. Cost: Medium. Implementation: Easy.
3. **Organize community clean-up drives** — Encourage residents, schools, and local organizations to participate in regular clean-up events. Cost: Low. Implementation: Easy.
4. **Start a recycling program** — Provide separate bins for recyclables and educate citizens on proper sorting. Cost: Medium. Implementation: Medium.
5. **Use smart waste bins** — Install bins with fill-level sensors to optimize collection routes and prevent overflow. Cost: High. Implementation: Difficult.

Use Case Comparison Table

The table below summarizes all five use cases side by side for quick reference and comparison. All five prompts have been tested with real LLM responses (see individual use case sections above for full outputs).

Use Case	Target User	Input	AI Process	Expected Output	Limitations	Possible Improvement	Tested ?
AI Email Assistant	Administrative staff across various government departments.	The email received from a citizen, along with the key information that should be included in the reply, such as processing timelines or confirmation that a request has been received.	The AI reads the citizen's email and the necessary information, then writes a professional email in the proper format.	A ready-to-send professional email with correct tone and no missing facts.	The AI may not know current policies unless they are provided, and it may respond too generally to legal or official matters, so a person must review it before sending.	Connect the AI to the department's latest policy documents so it can always provide correct, up-to-date information.	Yes
AI Report Summarizer	Senior officials and department heads in public-sector offices.	A long report, such as a quarterly budget report.	The AI reads the report and produces an executive summary with proper headings, sections, and page numbers.	A one-page summary with proper headings, sections, and page numbers.	The AI may not understand charts and tables and may misinterpret them if formatting is lost in plain text; formatting can affect accuracy.	If the AI is uncertain about any number, text, or table, it should explicitly flag this instead of guessing.	Yes
AI Meeting Notes Generator	Secretaries and office administrators.	Input can be rough meeting notes, a written transcript, or a transcript generated from a recorded online meeting.	The AI organizes the meeting transcript into structured minutes, covering topics discussed and decisions made. If a recording is provided, the audio is first converted to text using speech-to-text transcription, and the resulting transcript is then organized by the LLM into structured minutes.	A formatted minutes document with clear sections and a list of action items.	If speaker names are not mentioned in the transcript, the AI cannot verify who said what. It may also misinterpret jokes or informal conversation, and speech-to-text transcription may introduce errors caused by background noise.	Use meeting notes or transcripts with clearly labeled speaker names. The AI could also be connected to online meeting platforms such as Zoom so it can automatically generate transcripts.	Yes

Use Case	Target User	Input	AI Process	Expected Output	Limitations	Possible Improvement	Tested ?
AI Document Q&A; Assistant	Citizens and government staff.	A policy document and a specific question.	The AI searches the document and answers the question using only the information available within it.	A direct, accurate answer with a reference to where in the document it was found.	It may guess an answer if it is not present in the document, and very large documents may be difficult to process fully.	The AI should clearly state that a piece of information is not available in the document instead of guessing.	Yes
AI Idea Generator for Public Services	Policy planners and innovation teams in government departments.	A description of the problem that needs to be resolved, such as long queues at a passport office.	The AI generates multiple improvement ideas and explains how practical each one is.	A list of 5-7 useful ideas with simple notes about cost and imple mentation.	Ideas may be too generic. The AI does not know the actual budget, so it might suggest ideas that don't fit financially, or ideas that have already been tried.	Feed the AI historical data on past initiatives so it avoids repeating solutions that already failed.	Yes

Prompt Examples (Reference Summary)

The full tested prompt used for each use case is listed below for quick reference. See each use case section above for the corresponding actual tested output and evaluation note.

1. AI Email Assistant

“Write a formal reply email from the Ministry of Education to a student who is asking why their scholarship has not yet been granted, even though they applied a few months ago. Explain that the process takes time because the submitted documents must first be verified, and then the Ministry needs to receive confirmation from the student's university before the scholarship can be approved. Apologize for the delay, reassure the student that their application is being processed, and ask them to wait patiently. Keep the email under 150 words. Use a professional tone with a proper greeting, closing, and official email format.”

2. AI Report Summarizer

“Summarize this quarterly budget report. Organize the summary with headings for each department. Include the page number for every key statistic. Keep the summary under 400 words and end it with a short conclusion.”

3. AI Meeting Notes Generator

“Read this meeting transcript and convert it into formal meeting minutes. Include the meeting title (if available), date (if mentioned), attendees, topics discussed, key decisions, action items with the responsible person and deadline (if mentioned), and a summary of the important points discussed by each attendee. Use professional English, organize the information under clear headings, and present the important points in bullet form. Do not add any information that is not mentioned in the transcript.”

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“Read this attached document and answer this question: What was the total utilization rate this quarter, and what are the department's recommendations? List the recommendations in bullet points.”

5. AI Idea Generator for Public Services

“Suggest 5 simple and practical ideas to improve waste collection and cleanliness in a city. For each idea, write one short explanation and mention whether it is low, medium, or high cost and whether it is easy or difficult to implement.”

Required Submission Checklist

• Concept note PDF — this document

• Use case table — included above

• Prompt examples — included above, with real tested outputs for all 5 use cases

• GitHub commit link — add the link here once uploaded: [PASTE YOUR GITHUB COMMIT LINK HERE]

Reminder: Before submitting, confirm the GitHub repository is clean — remove or replace any empty placeholder files (such as the empty `notebook` file flagged in previous feedback) with the correct content, then add the commit link above.

Expected Outcome: This concept note demonstrates the ability to identify useful, practical Generative AI opportunities within a public-sector context and explain them in a clear, professional, and evidence-based format, supported by real tested prompts and outputs.

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